

Lab 02 Applications of propositional logic

Objective

Solving exercises from the textbook in chapter 1.2-1.3

Current Lab Learning Outcomes (LLO)

By completion of the lab, the students should be able to:

Understand the truth values of the compound propositions, negating the compound proposition

Lab Requirements

Students allowed using their lecture notes in the lab in order to solve the exercises.

Lab Assessment

- 1- Divide students to groups and let them to solve the given example.
- 2- Discuss the answers with the groups and write on board the optimal solution.

Lab Description

1. Translate the given statement into propositional logic using the propositions provided
You cannot edit a protected Wikipedia entry unless you are an administrator. Express your answer in terms of e : "You can edit a protected Wikipedia entry" and a : "You are an administrator"
2. Express these system specifications using the propositions p "The message is scanned for viruses" and q "The message was sent from an unknown system" together with logical connectives (including negations).
 - a) "The message is scanned for viruses whenever the message was sent from an unknown system."
 - b) "The message was sent from an unknown system but it was not scanned for viruses."
 - c) "It is necessary to scan the message for viruses whenever it was sent from an unknown system."
 - d) "When a message is not sent from an unknown system it is not scanned for viruses."
3. Determine whether each of these conditional statements is true or false.
 - a) If $1 + 1 = 2$, then $2 + 2 = 5$. b) If $1 + 1 = 3$, then $2 + 2 = 4$.
 - c) If $1 + 1 = 3$, then $2 + 2 = 5$. d) If monkeys can fly, then $1 + 1 = 3$.
4. State the converse, contrapositive, and inverse of each of these conditional statements.
 - a) If it snows tonight, then I will stay at home. b) I go to the beach whenever it is a sunny summer day. c) When I stay up late, it is necessary that I sleep until noon.
5. Show that $(p \rightarrow r) \wedge (q \rightarrow r)$ and $(p \vee q) \rightarrow r$ are logically equivalent using truth table.
6. Show that $(p \rightarrow r) \vee (q \rightarrow r)$ and $(p \wedge q) \rightarrow r$ are logically equivalent using truth table
7. Show that $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$ is a tautology using truth table
8. Use De Morgan's laws to find the negation of each of the following statements.
 - a) Jan is rich and happy.
 - b) Carlos will bicycle or run tomorrow.
 - c) "If elephants can fly, then the Cubs will lose the World Series this year"
 - d) if I am not hungry then it is not noon

9. Are these system specifications consistent?

- a. “If the file system is not locked, then new messages will be queued. If the file system is not locked, then the system is functioning normally, and conversely. If new messages are not queued, then they will be sent to the message buffer. If the file system is not locked, then new messages will be sent to the message buffer. New messages will not be sent to the message buffer.”
- b. “The router can send packets to the edge system only if it supports the new address space.”. “For the router to support the new address space it is necessary that the latest software release be installed.”. “The router can send packets to the edge system if the latest software release is installed”. “The router does not support the new address space.”

10. Determine whether the compound propositions are satisfiable.

- a) $(p \vee \neg q) \wedge (\neg p \vee q) \wedge (\neg p \vee \neg q)$
- b) $(p \rightarrow q) \wedge (p \rightarrow \neg q) \wedge (\neg p \rightarrow q) \wedge (\neg p \rightarrow \neg q)$
- c) $(p \leftrightarrow q) \wedge (\neg p \leftrightarrow q)$